

Civics Group	Index Number	Name (use BLOCK LETTERS)	H2



ST. ANDREW'S JUNIOR COLLEGE
2011 JC2 Preliminary Examinations

H2 BIOLOGY

9648/2

Paper 2: Core (Mark Scheme)

Tuesday

16 September 2011

2 hours

Additional Materials: Answer Paper
 Cover Sheet for Section B

READ THESE INSTRUCTIONS FIRST

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use a soft pencil for any diagram, graph or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A (Structured Questions)

Answer **all seven** questions.

Write your answers in the spaces provided on the question paper.

Section B (Essay Question)

Answer **one** essay question.

Write your answers on the separate answer paper provided.

All working for numerical answers must be shown.

INFORMATION TO CANDIDATES

At the end of the examination,

1. Attach Section B answers to the **cover sheet** provided.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/11
2	/11
3	/8
4	/13
5	/12
6	/13
7	/12
Total	/80

This document consists of **20** printed pages.

[Turn over

Section A

Answer all questions.

1. Fig. 1.1 below shows how the amount of DNA in a cell of *Chrysanthemum makinoi* varies during periods P to T in the cell cycle.

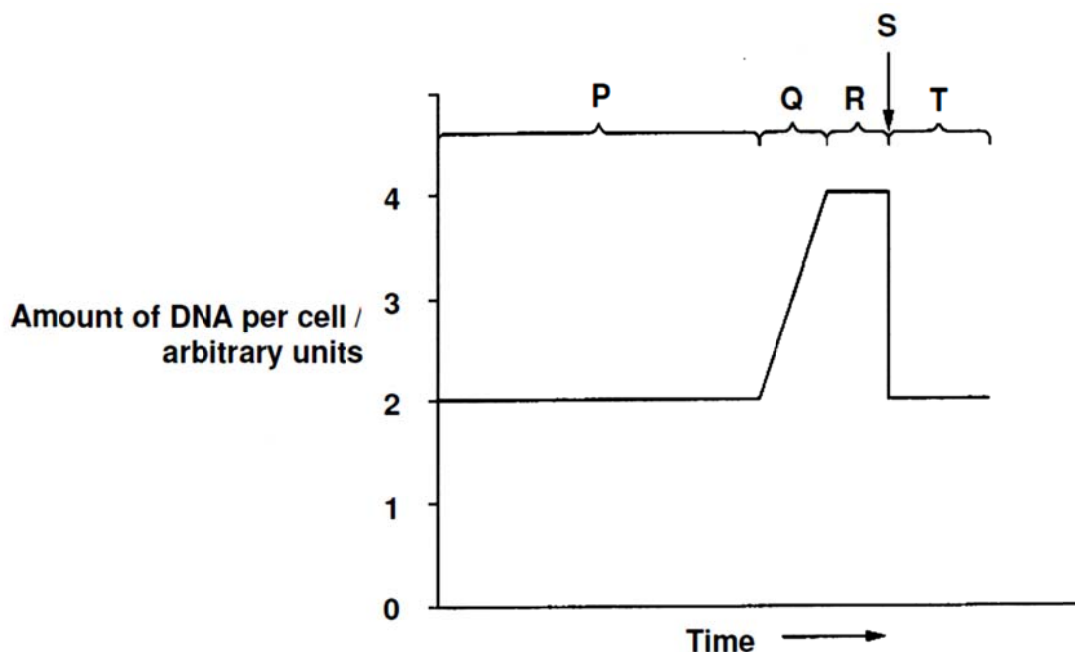


Figure 1.1

(a) Radioactive thymine was supplied to the cells of some growing *Chrysanthemum* tissue. With reference to Fig. 1.1, state precisely during which period of the cell cycle does radioactivity of the nuclei first increase. Explain your answer.

.....[2]

- 1 Period Q; corresponds to S phase (of Interphase);
- 2 {Semi-conservative DNA replication / radioactive thymine is incorporated into newly synthesized DNA strands}; resulted in doubling of DNA amount from 2 to 4 a.u.;

(b) At anaphase of mitosis, sister chromatids separate to become individual chromosomes. Identify the period in Fig. 1.1 during which anaphase occurs. Briefly explain your answer.

.....[3]

- 1 Period R;;
- AND
- 2 Separation of sister chromatids during anaphase occurs within the same cell;;
 - 3 DNA amount {remains unchanged at 4 a.u. / does not halved } prior to cytokinesis of the cell into two daughter cells;;

The first phase within interphase, from the end of the previous M phase until the beginning of DNA synthesis is called G₁ (G indicating *gap*). During this phase the biosynthetic activities of the cell, which had been considerably slowed down during M phase, resume at a high rate. This phase is marked by synthesis of various enzymes that are required in S phase, mainly those needed for DNA replication e.g. DNA polymerases known as Pol α .

(c) Describe briefly how the information on DNA is used to synthesize the RNA transcript for Pol α .

-[3]
- 1 RNA polymerase II ; General transcription factors ; Transcription initiation complex ; (**max 1**)
 - 2 bind to promoter sequence / TATA box; and unzips DNA helix ;
 - 3 adds complementary ribonucleotides; to the 3' end of growing RNA chain ;
 - 4 ref. termination of transcription; occurs 10 to 35 nucleotides downstream from by polyadenylation signal (of mRNA) ;

(d) In eukaryotes, the RNA transcript synthesized after transcription is not suitable for immediate translation by the ribosomes. Describe the eukaryotic processing of pre-mRNA by snRNPs.

-[3]
- 1 ref. spliceosomes = snRNPs + other proteins;
 - 2 recognition of splice sites ; cleaving at 5' end of intron;
 - 3 joins to branch point sequenc ; forms lariat structure;
 - 4 cleaving at 3' end of intron to release intron; ligation of exons together;

[Q1 Total: 11]

2.

(a) Describe the structural features of the influenza genome.

.....[2]

- 1 Single-stranded RNA;
- 2 Negative-sense RNA;
- 3 8 segments;
- 4 Each RNA segment associated with capsid proteins;

(b) Relate the structure of the influenza virus to its mechanism of penetration and release from the host.

.....[3]

Penetration

- 1 Haemagglutinin on viral particle recognise; and bind to sialic acid residues on host cells;
- 2 Invagination of cell membrane / formation of vesicle / endosome which encloses viral particle / virus enters by (receptor-mediated) endocytosis;;

Release

- 3 Viral neuraminidase cleaves sialic acid on viral envelope; to allow release of virus by budding;

Pandemics are epidemics of infectious diseases that are spreading through human populations across a large region or even worldwide, usually costing many people to lose their lives e.g. the Spanish flu in 1918 caused the death of at least 50 million people worldwide. Throughout history, there had been a number of pandemics, such as smallpox (viral disease) or tuberculosis (bacterial disease).

Fig. 2.1 highlights the significant dates in human influenza history.

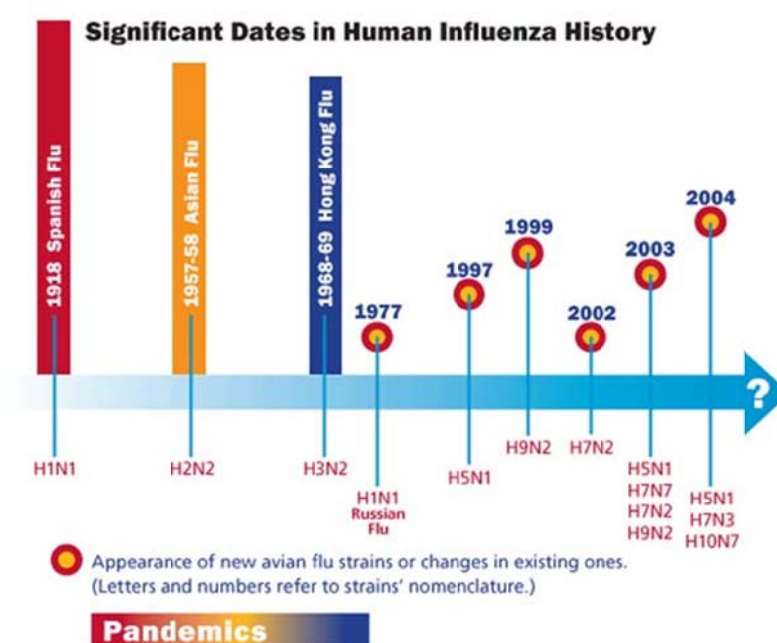


Figure 2.1

(c) Explain how the influenza pandemics arise.

-[4]
- 1 antigenic shift / reassortment of RNA segments;; **REJECT** antigenic drift
 - 2 occurs when different viral strains/subtypes infect the same cell;; **REJECT** different viruses
 - 3 leads to viruses with new combinations of haemagglutinin and neuraminidase genes;;
 - 4 Production of new haemagglutinin / neuraminidase antigens on viral envelope;;
 - 5 (Individuals have) no protection from antibodies produced from exposure to previous strains;;
- max 3**

The Acquired Immune Deficiency Syndrome (AIDS) pandemic is a widespread disease caused by the Human Immunodeficiency Virus (HIV). Since AIDS was first recognized in 1981, it has led to the deaths of more than 25 million people, making it one.

(d) Other than the mechanism of penetration and release from the host cell, state one similarity and one difference between the reproductive cycle of the HIV and influenza viruses

.....[2]

Similarity

- 1 Both viruses utilize host cell replication machinery (for replication of genome);;
- 2 Both viruses utilize transcription and translation machinery (for synthesis of viral protein);;

REJECT “metabolic machineries”

Differences

HIV reproductive cycle	Influenza productive cycle
Reverse transcription of RNA genome in HIV;	No reverse transcription in influenza;
Integration viral genome (d/s cDNA) into host chromosome	No integration of viral genome into host chromosome;
Latency phase in HIV life cycle;	No latency period in influenza;
Proteolysis of viral polyprotein to form mature viral proteins;	No proteolysis;

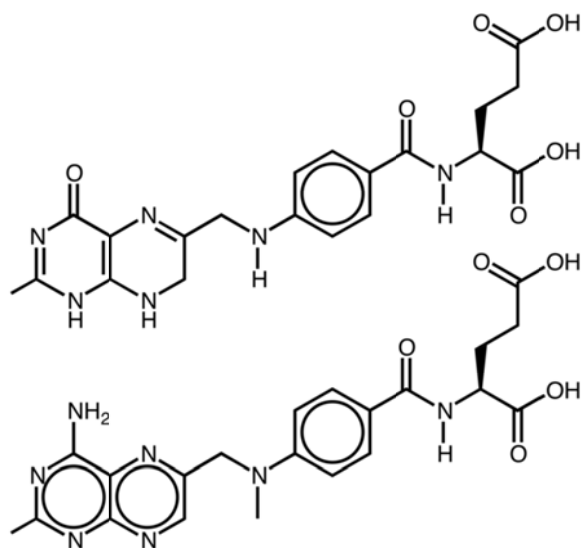
REJECT structural differences e.g. reverse transcriptase vs no reverse transcriptase, enveloped virus vs not enveloped virus

[Q2 Total: 11]

3.

(a) Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This drug slows down cell division by acting on one of the enzymes dihydrofolate reductase, in a metabolic pathway. This enzyme catalyses the conversion of dihydrofolic acid to tetrahydrofolic acid, that results in the formation of purines. The chemical structures of both dihydrofolic acid and Methotrexate are shown in Fig. 3.1.



3.1
Structures of
dihydrofolic acid (top)
and Methotrexate
(bottom).

Explain how the use of Methotrexate will slow down the mitotic cell cycle.

.....[3]

- 1 Methotrexate has a structural resemblance to dihydrofolic acid;;
- 2 Methotrexate is a competitive inhibitor / competes with the substrate for binding at the active site of the enzyme;;
- 3 Inhibition leads to less purines formed, and therefore less nucleotides synthesized; leading to decreased DNA replication;

(b) Binary fission differs significantly from the mitosis of eukaryotic cells. Describe the process of binary fission in bacterial cells.

.....[5]

- 1 the (single double-stranded) DNA chromosome attaches to the plasma membrane; at the origin of replication;
- 2 DNA chromosome replicates via bidirectional replication; and the newly synthesized DNA's origin of replication is also attached to the plasma membrane; ref. semi-conservative replication;
- 3 Bacterial cell elongates; and the two bacterial chromosomes origin of replication ends up in opposite poles of the cell;
- 4 The plasma membrane invaginates; pinching off the two halves of the original cell;
- 5 New cell wall material is synthesized;

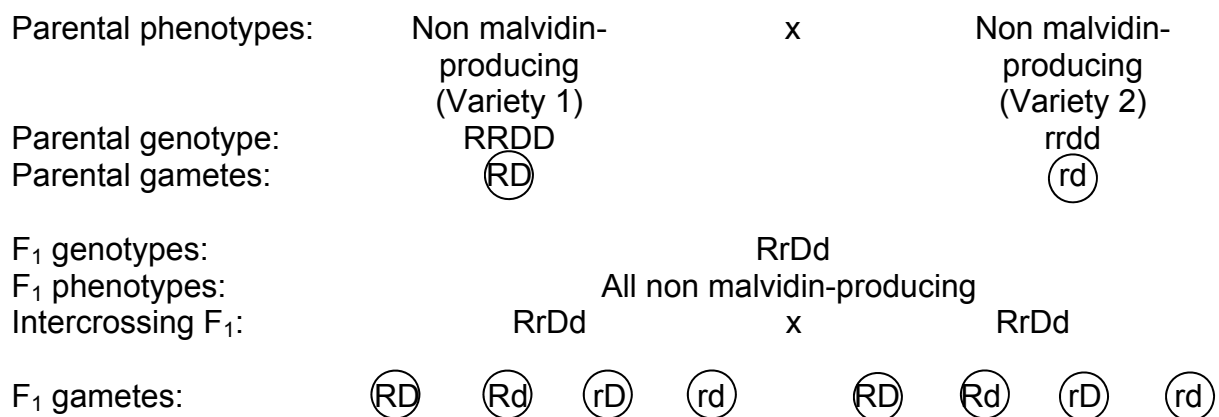
[Q3 Total: 8]

4. The chemical malvidin is one of the anthocyanidins responsible for the blue pigment found in the flowers of the *Primula* plants. Production of malvidin is controlled by two independent-assorting genes.

The dominant allele **R** codes for the production of malvidin. Plants with the dominant allele **D** results in the absence of malvidin production, even if they also carry the dominant allele **R**.

Non malvidin-producing Variety 1 has the genotype **RRDD** whilst non malvidin-producing plant Variety 2 has the genotype **rrdd**. The two varieties are crossed and the F_1 offspring intercrossed to give the F_2 generation.

(a) (i) Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the F_1 and F_2 generations of the two crosses. Give the ratio of phenotypes in the F_2 generation. [6]



Punnett square showing F_2 genotypes:

Gametes	RD	Rd	rD	rd
RD	Δ RRDD	Δ RRDd	Δ RrDD	Δ RrDd
Rd	Δ RRDd	# RRdd	Δ RrDd	# Rrdd
rD	Δ RrDD	Δ RrDd	Δ rrDD	Δ rrDd
rd	Δ RrDd	# Rrdd	Δ rrDd	Δ rrdd

Legend

Δ Non malvidin-producing

Malvidin-producing

F_2 phenotypes: Non malvidin-producing Malvidin-producing

F_2 phenotypic ratio: 13 3

- 1 F₁ genotypes – RrDd
- 2 F₁ phenotype – Non-malvidin producing / non-blue REJECT “white”
- 3 F₁ gametes – $\textcircled{RD}$ $\textcircled{Rd}$ $\textcircled{rD}$ $\textcircled{rd}$ (Gametes must be circled)
- 4 Punnett square showing all F₂ genotypes
- 5 corresponds genotypes to phenotypes / legend for Punnett square
- 6 F₂ phenotypic ratio – 13:3

(ii) Suggest the molecular basis for this form of epistasis.

-[2]
- 1 The dominant allele R produces an enzyme/protein involved in the biosynthesis of malvidin;;
 - 2 The dominant allele D produces an enzyme that breaks down malvidin / inhibitor of the enzyme involved in the biosynthesis of malvidin;;

(b) Pure breeding fruit flies, *Drosophila melanogaster*, with grey body and normal wings were crossed with pure breeding black body, vestigial wings. All the F₁ fruit flies had grey body and normal wings. Test cross of the F₁ fruit flies produced the following results:

grey body, normal wings	178
grey body, vestigial wings	132
black body, normal wings	142
black body, vestigial wings	160

- (i) Calculate the value of χ^2 applicable to these data. (Workings need not be shown).
[1]

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad v = c - 1$$

where Σ = ‘sum of...’ O = observed ‘value’
 v = degrees of freedom E = expected ‘value’
 c = number of classes

A chi-squared table.

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

<i>Classes</i>	<i>Observed number (O)</i>	<i>Expected ratio</i>	<i>Expected number (E)</i>
grey body, normal wings	178	1	153
grey body, vestigial wings	132	1	153
black body, normal wings	142	1	153
black body, vestigial wings	160	1	153
Total	612	4	612

No. of degrees of freedom

$$= c - 1$$

$$= 4 - 1 = \underline{3}$$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$= \frac{(178 - 153)^2}{153} + \frac{(132 - 153)^2}{153} + \frac{(142 - 153)^2}{153} + \frac{(160 - 153)^2}{153}$$

$$= \underline{8.08}$$

Examiners' comments

Round to 2 decimal places. Follow the decimal placing of the chi-squared table provided.

(ii) Use the calculated value of χ^2 and the table of probabilities provided to find the probability of the results of the test cross departing significantly by chance from the expected ratio.

Probability =[1]

0.02 – 0.05 or 2% – 5%

(ii) State what conclusions may be drawn from the probability found in (b)(ii)?

.....[3]

Award pts 1 and 2 only if probability found in (b)(ii) is correct

- 1 Since p value is smaller than 0.05, (there is only a small probability that the difference between observed and expected results is due to chance);
- 2 Difference between expected results and actual results is not due to chance
/There is significant difference between expected results and actual results;
- 3 No independent assortment of genes;;
- 4 Partially linkage of genes for body colour and wing shape
/ ref. linkage with crossing over of genes for body colour and wing shape/length
/ genes for body colour and wing shape are located on the same chromosome,
but crossing over occasionally breaks the linkage;;

[Q4 Total: 13]

5.

(a) An investigation into the production of ATP by mitochondria was carried out. ADP, phosphate, excess substrate and oxygen were added to a suspension of isolated mitochondria.

(i) Suggest the substrate used for this investigation.

.....[1]

- 1 pyruvate / acetyl coA / succinate etc.
/ any suitable substrate from the link reaction or Krebs cycle;;

(ii) Explain why the concentration of oxygen and amount of ADP fell during the investigation.

.....[2]

- 1 ADP and phosphate forms ATP / phosphorylation of ADP to form ATP;;
- 2 oxygen is the final electron acceptor to form water;;

(iii) A further investigation was carried out to find out the effect of three inhibitors, **A**, **B** and **C**, on the electron transport chain in these mitochondria. In each of three experiments, a different inhibitor was added. The table below shows the state of the electron carriers, **W–Z**, after the addition of inhibitor.

Inhibitor added	Electron carrier			
	W	X	Y	Z
A	oxidised	reduced	reduced	oxidised
B	oxidised	oxidised	reduced	oxidised
C	reduced	reduced	reduced	oxidised

Give the order of the electron carriers in this electron transport chain.

Order [1]

- 1 Y X W Z

(iv) Lactate dehydrogenase is responsible for the conversion of pyruvate to lactate by the enzyme. What would be the effect of inhibiting lactate dehydrogenase in a mammalian cell under anaerobic conditions?

.....[2]

- 1 NAD^+ cannot be regenerated;;
- 2 Glycolysis decreases / No ATP generation;;

(v) Plants can also respire anaerobically, but only for a short period of time. Explain why.

.....[1]

1 end-product of alcoholic respiration is ethanol; which is toxic to plants;

(b) Fig. 5.1 below shows the results of an experiment in photosynthesis using a unicellular green alga and the radioactive isotope ^{14}C to label carbon dioxide. The algae were actively photosynthesizing before the start of the experiment.

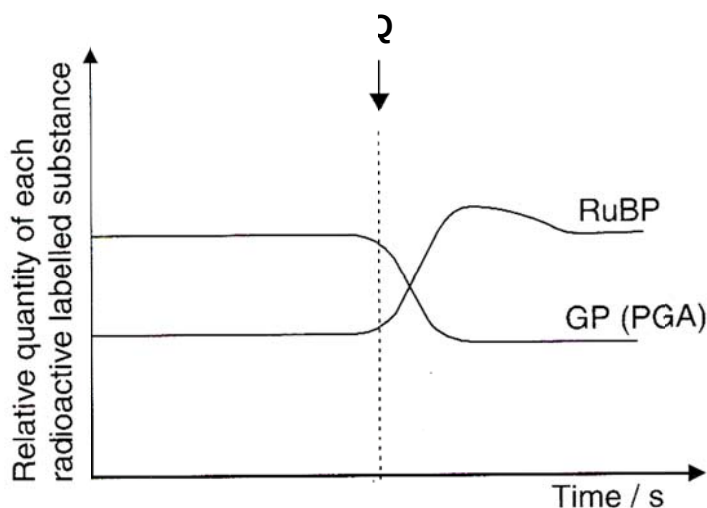


Figure 5.1

(i) State and explain what experimental condition was changed at Time Q.

.....[3]

- 1 The concentration of CO_2 was reduced;; **REJECT** "removed"
- 2 RuBP concentration increases due to reduced carbon fixation with CO_2 ; regeneration of RuBP from PGAL;
- 3 PGA concentration decreases due to conversion to PGAL; reduced carbon fixation with CO_2 ;

(ii) State two differences between the Krebs cycle and the Calvin cycle.

$\frac{1}{2}$ mark for correct comparison; $\frac{1}{2}$ mark for correct details

Krebs Cycle	Calvin Cycle
1 Occurs in <u>matrix of mitochondrion</u>	Occurs in <u>stroma of chloroplast</u>
2 Electron carrier = <u>NAD^+</u>	Electron carrier = <u>NADP^+</u>
3 Carbon dioxide <u>released</u> by oxidative decarboxylation	Carbon dioxide is <u>fixed</u> by Rubisco.
4 <u>ATP synthesised</u> by substrate level phosphorylation	<u>ATP used</u> in the reduction of PGA to PGAL, and the regeneration of RuBP

[2]

[Qn 5 - Total: 12]

6. Fig. 6.1 shows the changes in membrane potential at one point on an axon when an action potential is generated.

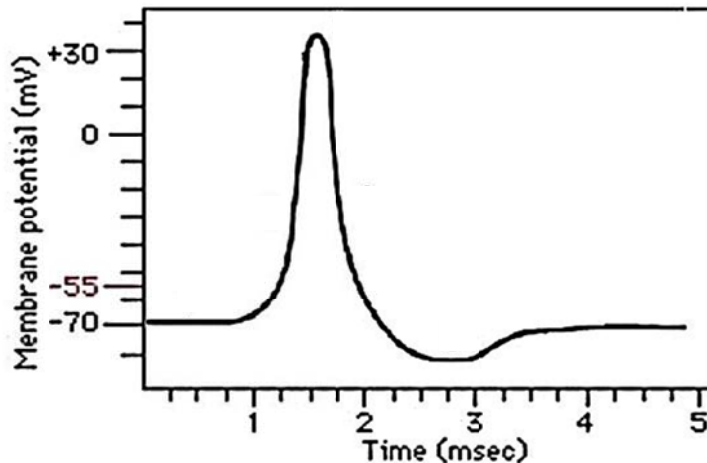


Figure 6.1

(a) Based on your knowledge of the fluid mosaic model and properties of membranes, account for the shape of the graph between 1.5 msec and 2 msec.

.....[5]

- 1 phospholipids are impermeable to ions; due to the hydrophobic core;
- 2 ions can only move in or out of neurones by facilitated diffusion/active transport; with the help of membrane proteins;
- 3 between 1.5msec to 2msec, repolarisation occurs; whereby membrane potential decreases from +40mV to -70mV; (ACCEPTABLE range +30mV to +40mV)
- 4 voltage-gated Na^+ ion channels (in the membrane) are closed; decreasing/stopping diffusion of Na^+ ions into neurone;
- 5 voltage-gated K^+ ion channels (in the membrane) open; increasing diffusion of K^+ ions out of neurone;

(b) Explain why a myelinated axon uses less energy than an unmyelinated axon of the same diameter to transmit a nerve impulse.

.....[1]

- 1 Na^+ (actively) pumped out only at nodes of Ranvier in myelinated axon / Na^+ (actively) pumped out along entire length of axon in non-myelinated axon;;

Fig. 6.2 shows the generation of action potentials against time at the pre-synaptic terminal and a post-synaptic membrane.

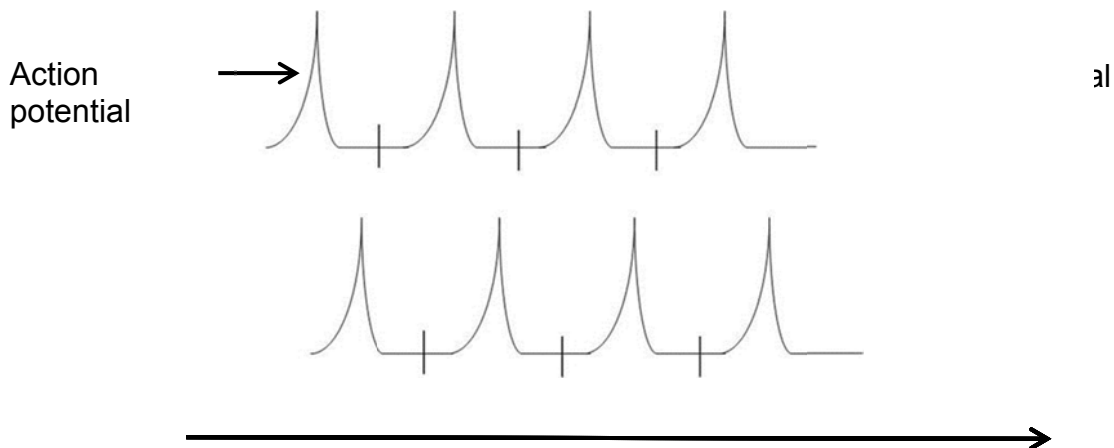


Figure 6.2

(c) With reference to the figure, account for the difference in the time of action potential generation.

-[2]
- 1 ref. time lag between action potential generation in the pre-synaptic and post-synaptic membranes
/ post-synaptic membrane gets depolarized later
/ action potential is generated in the post-synaptic membrane only after action potential is generated at pre-synaptic terminal;;
 - 2 time required for release from the synaptic vesicles; and diffusion of the neurotransmitter across the synaptic cleft;

(d) A peptide has been developed in the pharmaceutical industry as a treatment for severe chronic pain. It is the synthetic equivalent of the venom of the marine snail (*Conus magus*), which induces paralysis in fishes. This peptide is found to affect voltage-sensitive calcium channels of neurones.

Describe how this peptide can result in pain relief.

-[5]
- 1 blocks voltage-gated calcium channels; Ca^{2+} ions cannot enter the pre-synaptic terminal;
 - 2 prevents synaptic vesicles from migrating towards pre-synaptic membrane; and fusing with pre-synaptic membrane;
 - 3 prevents neurotransmitter/acetylcholine from being released into the synaptic cleft/ prevents exocytosis ; no diffusion of neurotransmitter across synaptic cleft;
 - 4 no binding of neurotransmitter to post- synaptic membrane receptors; no EPSP generated/ no depolarisation of post-synaptic membrane;
 - 5 transmission of pain to the brain is inhibited;;

[Q6 Total: 13]

7.

Lake Tanganyika is the deepest lake in Africa and the second largest lake by volume in the world. Scientists estimate the age of the lake to be between 9-12 million years old. Lake Tanganyika contains 3 major basins which formed separately and joined when their water rose over time.

The lake holds at least 250 species of cichlid fish and 150 non-cichlid species, which are highly diverse in terms of morphology, behaviour, and ecology. Almost all (98%) of the Tanganyikan cichlid species are endemic¹ to the lake.

Among the many species of cichlids, seven species in the genus *Perissodus* are known for their unique feeding habit of scale eating with specialized feeding morphology and behaviour.

Fig 7.1 shows the phylogenetic tree constructed based on the feeding habits (A) and the habitat depth and oral tooth shape (B) of nine species of *Perissodini*.

¹ Endemic- the ecological state of being unique to a defined geographic location.

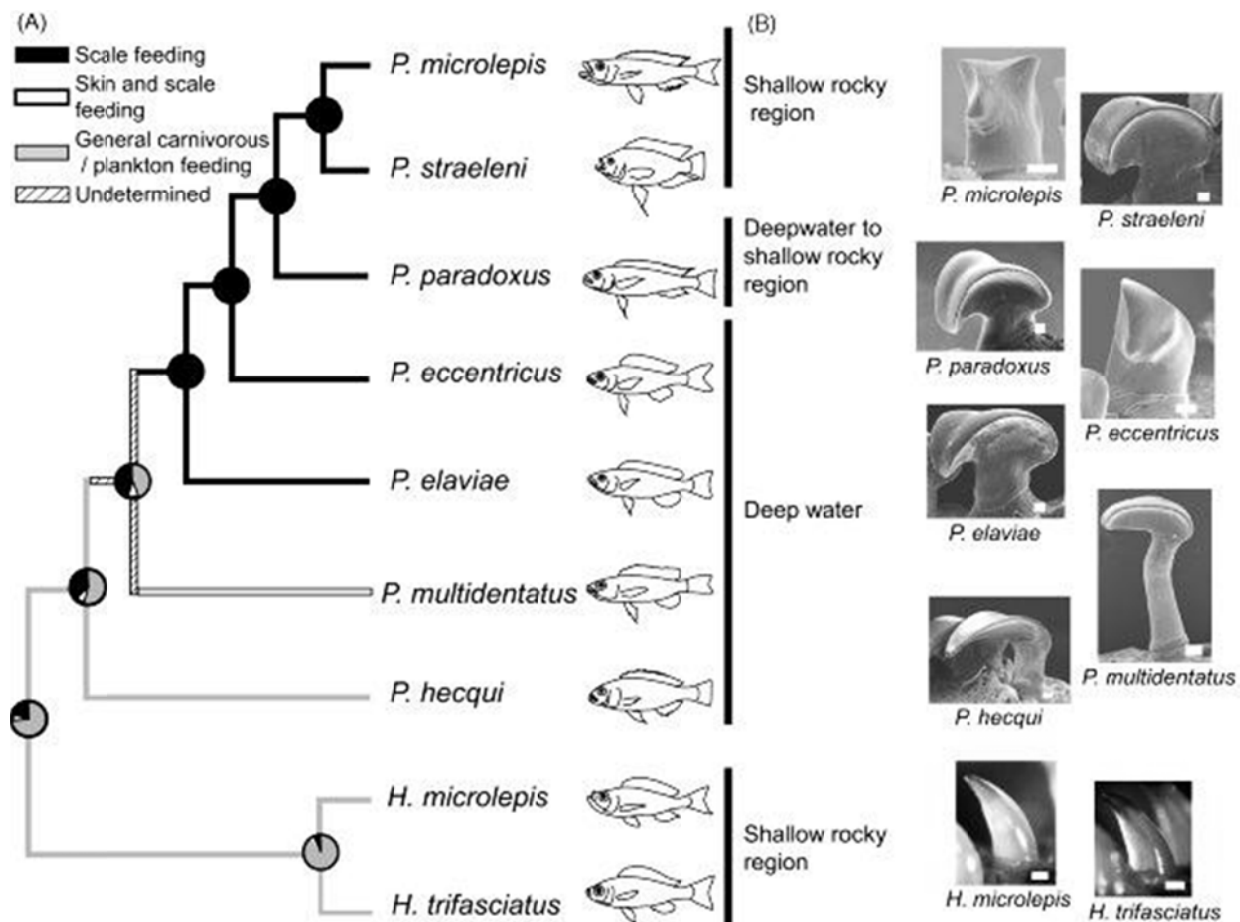


Figure 7.1

(a) Explain how natural selection could have caused the diversity of oral tooth shape in the different cichlid species in Lake Tanganyika.

-[4]
- 1 Genetic variations in the oral tooth shape exist; within the one or few species of cichlids due to mutation;
 - 2 Different selection pressures / different ecological niches available in different parts of the lake; such as different types of food available;
 - 3 Individuals with a selective advantage in the particular environment survived till reproductive age; and pass on their genes to their offspring;
/ ref. fish adapting to the specific selection pressures;
 - 4 ref. reproductive barriers/isolation; preventing gene flow between different populations of cichlids / leading to each population accumulating own genetic differences;
 - 5 ref. adaptive radiation / divergent evolution;

(b) Suggest a reason why it may be more reliable to construct a phylogenetic tree of the nine species of *Perissodini* using molecular data.

-[1]
- 1 Quantifiable ;
protein, nucleic acid sequence data are precise and accurate and easy to quantify / convertible to numerical form for mathematical and statistical analysis;

OR

- 2 Objective;
based strictly on heritable material
/ can be easily described in an unambiguous manner
/ some morphological characteristics may be analogous / ref. convergent evolution;

(c) Based on the fossil record, *P. microlepis* and *P. straeleni* share a common ancestor about 50,000 years ago. Describe how the molecular clock can be used to verify this information.

-[2]
- 1 Neutral mutations; are fixed/accumulated at a relatively constant rate for any particular gene;
 - 2 By determining the number of mutations/substitutions/differences; in a common gene/amino acid sequence in each organism, we could calculate the length of time since divergence;

A most peculiar behavior is seen in the scale-eating cichlid, *P. microlepis*, which feeds exclusively on the scales of other cichlids to survive. Phenotypes of this cichlid species are either right-mouthed or left-mouthed.

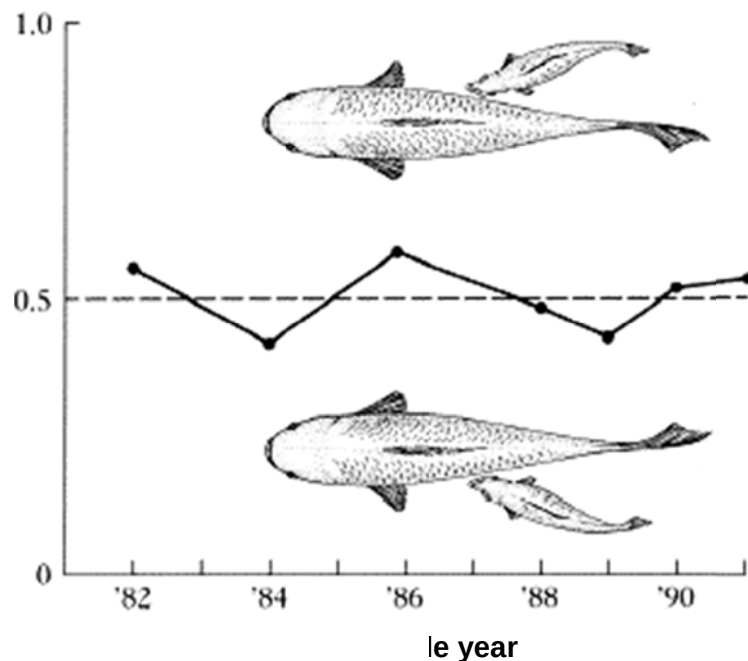


Figure 7.2

(d) The relative frequency of the left-mouthed fish and right-mouthed fish fluctuate around a mean of 0.5 (see Fig 7.2) and the presence of both phenotypes appears to be in an evolutionary stable state. Suggest an explanation for this phenomenon.

-[3]
- 1 ref. frequency-dependent selection;
 - 2 when a particular phenotype (i.e. left-mouthed fish) becomes more common in the population; prey fishes that are i.e. right side dominant, are better able to guard against the left-mouthed fish;
 - 3 Minority phenotype has a selective advantage compared to the majority phenotype and survive till reproductive age; pass on their genes to their offspring;
 - 4 Increase in frequency of minority phenotype;

(e) After a major drought, there was a sharp increase in the number of left-mouthed fish which lasted for a long period of time. Explain why.

-[2]
- 1 bottleneck effect; resulting in small population of survivors;
 - 2 loss of genetic variation through genetic drift; alleles coding for left-mouthed phenotype are over-represented
/ alleles coding for right-mouthed phenotype may be lost;

[Q7 Total: 12]

Section B

Answer **one** question.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections (a), (b) etc., as indicated in the question.

- 8** **(a)** Describe the structure and functions of the centrioles, nucleolus and smooth endoplasmic reticulum in the cell. [8]
- (b)** Contrast triglycerides and phospholipids. [4]
- (c)** Describe the main features of collagen that contribute to its tensile strength [8]

[Total: 20]

- 9** **(a)** Compare the structure and organisation of prokaryotic and eukaryotic chromosome. [6]
- (b)** Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [8]
- (c)** Describe the development of cancer as a multi-step process. [6]

[Total: 20]

8(a) Describe the structure and functions of the centrioles, nucleolus and smooth endoplasmic reticulum in the cell.

[8]

Centrioles

- 1 rod-like structures / cylindrical ;
- 2 positioned at right angles to each other ;
- 3 exists as a pair ;
- 4 consists of 9 triplets of microtubules;
- 5 Fn = organisation (REJECT formation) of spindle fibres during nuclear division;
- 6 important for separation of chromosomes / chromatids during meiosis /mitosis

Nucleolus

- 7 densely stained (granules and fibres) ;
- 8 observable in a non-dividing cell ;
- 9 one of more nucleoli may be present within nucleoplasm ;
- 10 Fn = site of transcription of rRNA genes / synthesis of rRNA ;
- 11 important for formation of ribosomal subunits involved in translation / protein synthesis / component of ribosomes

Smooth endoplasmic reticulum

- 12 consists of a network of fine tubules; REJECT “flattened sacs”
- 13 appears smooth due to the lack of ribosomes ;
- 14 Fn = synthesis of lipids ;
- 15 required by the cell as structural component of membrane, e.g. phospholipids
/ required by the cell as components of cell-cell recognition or cell-cell adhesion,
e.g. glycolipids
/ synthesis of hormones/steroids
- 16 Fn = detoxification of drugs and poisons ;
- 17 prevents built-up of toxic materials which may harm the cell ;
- 18 Fn = storage of Ca^{2+} ions (in muscle cells) ;
- 19 required for muscle contraction as Ca^{2+} functions as a second messenger ;

8(b) Contrast triglycerides and phospholipids.

[4]

	Triglycerides	Phospholipids
Structure	Consist of <u>three</u> fatty acids covalently bonded to a glycerol molecule ; No phosphate groups attached ; Do not have other groups attached to glycerol besides fatty acids ;	Consist of <u>two</u> fatty acids covalently bonded to a glycerol molecule ; Phosphate group attached to glycerol ; May have other groups attached to the phosphate group ; e.g. nitrogen-containing base choline
Properties	Hydrophobic / insoluble in water due to presence of hydrocarbon chains ;	Amphipathic due to presence of both hydrophilic and hydrophobic groups ;
Functions	Used for storage due to the lack of osmotic effect ; Used as a respiratory substrate due to its high energy per unit mass content compared to other substrates ; Required for the solubility of some fat-soluble vitamins due to their hydrophobic nature ; Source of metabolic water in some desert animals ; Provide aquatic animals with buoyancy ; Protection of delicate internal organs e.g. kidneys; Insulation to prevent excessive loss of heat in e.g. polar bears; max 1	Structural component of membranes in cells ; Involved in compartmentalisation for specialized conditions to be maintained in cells e.g. proton gradient in chloroplast, mitochondria; max 1

8(c) Relate the structural organization of collagen to its function.**[8]**

- 1 each polypeptide chain is 1,000 amino acid residues long
/ large molecular size of the tropocollagen molecule
- 2 rich in hydrophobic amino acids
- 3 ref. glycine and proline
- 4 collagen is insoluble in water

Intramolecular bonds (between the 3 polypeptides)

- 5 polypeptide chains cross-linked to each other
/ intramolecular bonds (between the 3 polypeptides)
- 6 triple helix / 3 polypeptide chains form helical structure
- 7 called tropocollagen
- 8 ref. hydrogen bonds
- 9 between the –NH groups (of glycine residues) on each strand and –CO groups on the other two strands
- 10 hydrogen bonds involving hydroxyproline to stabilise structure
- 11 small glycine residue fits into the crowded centre of the triple helix

Intermolecular bonds (between tropocollagen molecules)

- 12 tropocollagen molecules are cross-linked to each other
/ intermolecular bonding between tropocollagen molecules
- 13 covalent bonds involving lysine and hydroxylysine
- 14 to form collagen fibrils
- 15 which further assemble to form collagen fibres
- 16 staggered arrangement of tropocollagen in parallel chains;
- 17 no weak areas/ more difficult to break;
- 18 great tensile strength
- 19 resists tears and stretching
- 20 component of connective tissues, cartilage and skin

9(a) Compare the structure and organisation of prokaryotic and eukaryotic chromosome.

[6]

For every comparison:

½ mark for correct *comparison*

½ mark for correct information

	Prokaryotic chromosome	Eukaryotic chromosome
1. Genome size	Smaller genomes, 0.6 Mb to 10 Mb	Large genomes, being less than 10 Mb – 100,000 Mb
2. Gene length	Shorter gene sequences / more compact genetic organisation	Longer gene sequences / presence of more intergenic spaces
3. Chromosome structure	Circular DNA molecule.	Linear DNA, each contained in a different chromosome.
4. Packaging of DNA	Does not form chromatin. (Prokaryotic DNA is organized into a DNA-protein complex called the nucleoid.	Eukaryotic DNA is complexed with histones and other proteins to form chromatin.
5. Introns	Coding sequence proceeds from start to finish without interruption by introns.	Presence of introns within genes.
6. Regulatory sequences	Small amount of non-coding DNA consists mainly of regulatory sequences, such as promoters)	More complex regulatory sequences (eg. enhancers and silencers)
Others (max 3)		
7. Chromosome number	Single chromosome / Haploid	Many chromosomes / Diploid or polyploid
8. Location	Chromosome found in the nucleoid region of the cytoplasm	Chromosomes are enclosed within a double-membrane bound nucleus
9. Repetitive sequences	Few repetitive DNA sequences.	Many highly repetitive DNA sequences
10. Coding and non-coding DNA	Most of DNA are coding sequences (codes for protein, tRNA, or rRNA.	Most of DNA are non-coding.
11. Presence and absence of operons	Two or more genes may be expressed and regulated as a unit (genomes arranged in operons / polycistronic genes)	Absence of operons / monocistronic genes
12. Origins of replication	One	Many
13. Telomeres	Absent	Present
14. Centromere	Absent	Present

9(b) Outline the differences between prokaryotic control of gene expression with the eukaryotic model. [8]

For every comparison:

½ mark for correct *comparison*

½ mark for correct information

Chromosomal Level (max 2)

Prokaryotes	Eukaryotes
1. Prokaryotic DNA not organised into chromatin / not associated with histones	Eukaryotic DNA is complexed with histones and other proteins to form chromatin / associated with histones
2. DNA and histone modification cannot occur	DNA and histone <u>modification</u> can occur, resulting in conversion between euchromatin and heterochromatin
3. DNA sequences, eg. promoters and operators, serve as the on/off switch	Structure of chromatin – euchromatin, ready to be transcribed, or heterochromatin and not available – is the major <u>on/off switch</u> for gene regulation / Ease of transcription – DNA made accessible to RNA polymerase and other regulatory proteins

Transcriptional Level (max 2)

Prokaryotes	Eukaryotes
4. One RNA polymerase consisting of five subunits / All RNAs synthesized by the same RNA polymerase;	Three different RNA polymerases, each containing 10 or more subunits; / Three different classes of RNA each synthesized by a different RNA polymerase (i.e. mRNA, tRNA, rRNA)
5. Simple regulatory sequence: Transcriptional regulatory protein / Regulator protein binds to DNA-binding sites upstream of the cluster of structural genes to regulate initiation of transcription.	Complex regulatory sequence: More extensive interaction between upstream DNA sequences and protein factors involved to stimulate and initiate transcription. In addition to promoters, enhancers and silencers control rate of transcription.
6. Related genes are transcribed together as operons / only 1 promoter / polycistronic mRNA	No operon / each gene has own promoter / monocistronic mRNA

Post-transcriptional Level (max 2)

Prokaryotes	Eukaryotes
7. Translation is often coupled to transcription / Transcription and translation take place in the same cellular compartment simultaneously.	No direct coupling of transcription and translation / mRNA must pass across nuclear envelope before translation in the cytoplasm. RNA transcript is not free to associate with ribosomes prior to the completion of transcription.
8. Primary transcripts are the actual mRNAs / no post-transcriptional modification	Primary transcripts undergo processing to produce mature mRNAs – methylated guanosine cap at the 5' end, poly-A tail at the 3' end, splicing
9. Lower stability of transcript / degradation within seconds or minutes / mRNAs shorter half life to rapidly respond to environmental changes	Higher stability of transcript / prevent transcript degradation / mRNAs longer half-life remaining much longer to orchestrate protein synthesis prior to their degradation by nucleases in the cell

Translational level (max 2)

Prokaryotes	Eukaryotes
10. Control at this level is unlikely; due to simultaneous transcription and translation	Control at translational level: phosphorylation of ribosomal translation initiation factors / negative translational control through regulatory proteins / cytoplasmic elongation of poly (A) tails / mRNA degradation / RNA interference and microRNA
11. mRNAs have multiple ribosome binding sites / direct the synthesis of several different polypeptides	mRNAs have only one start site / direct synthesis of only one kind of polypeptide

Post-translational Level (max 2)

Prokaryotes	Eukaryotes
12. no/minimal post-translational modifications occur	Post-translational modifications determine the functional abilities of the protein
13.	Proteolysis: <u>Processing</u> eukaryotic polypeptides to yield <u>functional</u> protein molecules e.g. cleavage of pro-insulin to form the active insulin hormone
14.	Chemical modification of proteins to yield functional protein molecules
15.	Phosphorylation of proteins to increase or decrease its function
16.	Transportation of proteins to target destinations in the cell where it functions is mediated by signal sequences at N-terminus of some proteins. Once transported to destination, signal sequence is enzymatically removed from the proteins
17.	Ubiquitination marks protein for degradation, ref to ubiquitin & proteasome

9(c) Describe the development of cancer as a multi-step process.**[6]**

- 1 loss of function mutation of tumour suppressor gene
- 2 mutant tumour suppressor alleles are usually recessive
/ mutations must knock out both alleles in a cell's genome to produce abnormal protein
- 3 loss of arrest of cell division to allow more time for the cell to repair its DNA / loss of ability for DNA repair / loss of apoptosis
- 4 gain of function mutation of proto-oncogene
- 5 oncogenes behave as dominant alleles
/ only need one mutated allele to produce abnormal protein
- 6 lead to stimulation of the cell cycle / cell keeps dividing
- 7 accumulation of many mutations
- 8 activation of telomerase gene
- 9 telomerase enzyme prevents the shortening of the chromosome ends / cell can continue to divide indefinitely
- 10 angiogenesis / formation of new network of blood vessels to the cancer cells
- 11 blood vessels provide the cancer cells oxygen and nutrients for growth and to remove any waste products
- 12 loss of contact inhibition / density-dependence (and cells do not stop dividing)
- 13 loss of ability to differentiate
- 14 formation of benign tumour
- 15 cells in the tumour no longer exhibit anchorage dependence / loss of cell adhesion
- 16 metastasis / break loose and may enter the bloodstream / invade other tissues form secondary tumors
- 17 tumour is considered malignant